$4^3 = 4 \cdot 4 \cdot 4 = 64$

$25^0 = 1$

$(-9)^0 = 1$

$-2014^0 = 1$

$\left(-\frac{2}{5}\right)^{-1} = -\frac{5}{2}$

$\left(\frac{3}{4}\right)^{-2} = \frac{16}{9}$

$(-2)^{-5} = \left(-\frac{1}{2}\right)^5 = \frac{1}{32}$

$-\left[\left(-\frac{1}{2}\right)^{-1}\right]^4 = -\left(-\frac{1}{2}\right)^{-4} = -(-2)^4 = -2^4 = -16$

$-4^5 + 3.4^5 - 7.3^4 - 3.3^4 - 4^5 = 4^5.(-1+3-1) + 3^4.(-7-3)$

$= 4^5 - 10 \cdot 3^4$

$\frac{5^{80} + 5^{85} + 5^{90}}{5^{85} + 5^{80} + 5^{75}} = x$ ise $\frac{x}{125}$ ifadesinin eşiti nedir?

$x = \frac{5^{80} + 5^{85} + 5^{90}}{5^{85} + 5^{80} + 5^{75}} = \frac{5^{80}.(1 + 5^5 + 5^{10})}{5^{75}.(5^{10} + 5^5 + 1)} = \frac{5^{80}}{5^{75}} = 5^5$

Örnek: $\frac{14^n + 6^n + 2^n}{7^n + 3^n + 1}$ ifadesi neye eşittir?

$\frac{(7.2)^n + (3.2)^n + 2^n}{7^n + 3^n + 1} = \frac{7^n \cdot 2^n + 3^n \cdot 2^n + 2^n}{7^n + 3^n + 1} = \frac{2^n.(7^n + 3^n + 1)}{7^n + 3^n + 1} = 2^n$

Örnek: $(0{,}027)^{\frac{5}{3}}.10^5$ işleminin sonucu neye eşittir?

$(0{,}027)^{\frac{5}{3}} \cdot 10^5 = (27.10^{-3})^{\frac{5}{3}} \cdot 10^5 = (3^3)^{\frac{5}{3}} \cdot 10^5 = 3^5 \cdot 10^{-5} \cdot 10^5 = 3^5 \cdot 10^0 = 3^5$

Örnek: $\frac{-9^{-2}.25^{-1}}{125^{-1}.27^{-1}} = ?$

$\frac{-9^{-2} \cdot 25^{-1}}{125^{-1} \cdot 27^{-1}} = \frac{-(3^2)^{-2} \cdot (5^2)^{-1}}{(5^3)^{-1} \cdot (3^3)^{-1}} = -\frac{5^{3-2}}{3^{4-3}} = -\frac{5}{3}$

Örnek:

$\sqrt{0,64}+\sqrt{0,04}-\sqrt[3]{0,027}=$

$\sqrt{\frac{64}{100}}+\sqrt{\frac{4}{100}}-\sqrt[3]{\frac{27}{1000}} \Rightarrow \sqrt{\frac{8^2}{10^2}}+\sqrt{\frac{2^2}{10^2}}-\sqrt[3]{\frac{3^3}{10^3}} \Rightarrow \frac{8}{10}+\frac{2}{10}-\frac{3}{10}=\frac{7}{10}$

Örnek:

$\frac{\sqrt{75}-\sqrt{27}+\sqrt{12}}{\sqrt{108}+\sqrt{48}}=$ $\frac{\sqrt{5^2\cdot 3}-\sqrt{3^2\cdot 3}+\sqrt{2^2\cdot 3}}{\sqrt{6^2\cdot 3}+\sqrt{4^2\cdot 3}} \Rightarrow \frac{5\sqrt{3}-3\sqrt{3}+2\sqrt{3}}{6\sqrt{3}+4\sqrt{3}} \Rightarrow \frac{4\sqrt{3}}{10\sqrt{3}}=\frac{4}{10} \Rightarrow \frac{2}{5}$

Örnek: $\sqrt{2}.(\sqrt{18}+\sqrt{50}-\sqrt{32}+\sqrt{8})=$ $\sqrt{36}+\sqrt{100}-\sqrt{64}+\sqrt{16}$

$6+10-8+4=12$

Örnek: $\frac{\sqrt[5]{16}}{\sqrt[5]{\frac{1}{2}}}=$ $\sqrt[5]{\frac{16}{\frac{1}{2}}}=\sqrt[5]{32}=\sqrt[5]{2^5}=$

Örnek: $\sqrt{25-25x^2}-\sqrt{64-64x^2}=$ $\sqrt{25\cdot(1-x^2)}-\sqrt{64\cdot(1-x^2)}=5\sqrt{1-x^2}-8\sqrt{1-x^2}$

$=3\sqrt{1-x^2}$

Örnek: $\sqrt[5]{\sqrt{\sqrt{\sqrt[3]{\sqrt{2}}}}}=$ $5\cdot 2\cdot 2\cdot 3\cdot 2\sqrt{2}=120\sqrt{2}$

Örnek: $\sqrt[3]{2\sqrt[3]{4}}=\sqrt[3]{\sqrt[3]{2^{2\cdot 3}\cdot 4}}=$ $3\cdot 2\cdot 3\sqrt{2^6\cdot 2^2}=18\sqrt{2^8} \Rightarrow 9\sqrt{2^9}=2^{\frac{9}{2}}$

Örnek: a>0 olmak üzere; $\left(\sqrt[5]{a^2}\right)^4$ ifadesinin eşiti nedir?

$\left(\sqrt[5]{a^2}\right)^4=\sqrt[5]{a^8}=\sqrt[5]{a^5\cdot a^3}=a\cdot\sqrt[5]{a^3}$

Örnek: $\sqrt{(-4)^2}-\sqrt[3]{(-2)^3}+\sqrt{(-2)^4}+\sqrt[5]{3^5}=$

$|-4|+2+|-2|+3$

$+4 \qquad +2$

$=11$

Örnek: a<0 olmak üzere; $\sqrt{a^2}+\sqrt{(-a)^2}-\sqrt{4a^2}$ ifadesinin eşiti nedir?

$|a|+|-a|-|2a|$

$-a+(-a)+2a=0$

Örnek: a>0 ve b<0 ise, $\sqrt{(b-a)^2}-\sqrt{(2a-b)^2}$ ifadesinin eşiti nedir?

$\sqrt{(b-a)^2}-\sqrt{(2a-b)^2} = |b-a|-|2a-b|$

$= -(b-a)-(2a-b)$

$= -b+a-2a+b$

$= -a$

Örnek: $A=\sqrt{5}-\sqrt{3}$ ve $B=\sqrt{5}+\sqrt{3}$ olduğuna göre; $\frac{A}{B}+\frac{B}{A}+2$ ifadesi neye eşittir?

$$\frac{A}{B}+\frac{B}{A}+2=\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}+\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}+2=\frac{(\sqrt{5}-\sqrt{3})^2}{\sqrt{5^2}-\sqrt{3^2}}+\frac{(\sqrt{5}+\sqrt{3})^2}{\sqrt{5^2}-\sqrt{3^2}}+2$$

$(\sqrt{5}-\sqrt{3})$ $(\sqrt{5}+\sqrt{3})$

$$=\frac{5-2\sqrt{15}+3+5+2\sqrt{15}+3}{2}+2=8+2=10$$

Örnek: $\dfrac{x^{a+1}}{x^{a-2}} \cdot \left(\dfrac{x^{-a+2} \cdot x^{2a+3}}{x} \right)^{-1}$ ifadesinin eşiti nedir?

$\dfrac{x^{a+1}}{x^{a-2}} \cdot \left(\dfrac{x^{-a+2} \cdot x^{2a+3}}{x} \right)^{-1} = x^{a+1-a+2} \cdot \left(x^{-a+2+2a+3-1} \right)^{-1}$

$= x^3 \cdot (x^{a+4})^{-1} = x^3 \cdot x^{-a-4} = x^{-a-1}$

Örnek: $\left(\dfrac{a^x}{a^y} \right)^{x-y} \cdot \left(\dfrac{a^y}{a^x} \right)^{x-y}$ ifadesinin eşiti nedir?

$(a^{x-y})^{x-y} \cdot (a^{y-x})^{x-y} = (a^{x-y+y-x})^{x-y} = (a^0)^{x-y} = 1$

Örnek: $3^x = 2$ olduğuna göre $27^{x+1} - 81^x$ ifadesinin değeri kaçtır?

$27^{x+1} - 81^x \Rightarrow 3^{3x+3} - 3^{4x} \Rightarrow (3^x)^3 \cdot 3^3 - (3^x)^4 \Rightarrow 2^3 \cdot 3^3 - 2^4$

$2 \quad\quad 2$

$8 \quad 27$

$216 - 16 = 200$

Örnek: $x - 2 = y$ olduğuna göre $|x-y| + |y-x|$ ifadesi neye eşittir?

$2 = x - y$

$-2 = y - x$

$2 + |-2|$

$2 = 4$

Örnek: $x \in R$ ve $-2 < x < 4$ olduğuna göre $|x+2| + |x-4|$ ifadesinin eşiti nedir?

$-1, 0, 1, 2, 3$

$x+2 \quad -x+4$

$2+4 = 6$

$-2x+y \quad -2y-x$

Örnek: $x < 0 < y$ ise $\dfrac{|2x-y| - |2y-x|}{x+y}$ ifadesi neye eşittir?

$\dfrac{-x-y}{x+y} = -1$